Mathematical Memorandum IIIA
Regime-Dependent Generative Models for Active Inference

A Minimal Hybrid Extension Motivated by the Addictive Cycle

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Abstract

This memorandum develops the first part of a regime-dependent extension of Active Inference motivated by the phenomenology of addiction and recovery. The purpose is deliberately narrow. Standard variational free energy, expected free energy, preference distributions, policy precision, and Bayesian inference over policies are left unchanged. The proposed extension occurs at the level of the hidden generative model. A discrete regime variable is introduced to index distinct operating modes of the model, while a continuous latent state represents affective, cognitive, motivational, interoceptive, and precision-related variables. Emotional dysregulation is defined as an emergent functional of this continuous latent state, rather than as an additional state coordinate. Regime transitions are modeled by conditional probabilities depending on the current regime, latent state, dysregulation, and action or intervention. The central proposal is that a psychological regime should be interpreted mathematically as an operating mode of the generative model, not merely as a behavioral label. This memorandum provides the generative-model architecture on which subsequent memoranda can define planning, policy selection, and application to the five-regime addictive cycle.

Contents

1 Purpose and Scope

This memorandum is the first of three technical memoranda extending the explicit engineering formulation of Active Inference to a multiregime account of the addictive cycle. The present document has a restricted purpose: it defines the regime-dependent generative model. It does not rederive variational free energy or expected free energy, and it does not replace the core mathematics of Active Inference.

The companion conceptual paper proposed that addiction and recovery may be represented as a sequence of qualitatively distinct psychological operating regimes: Wise Mind or Recovery, Emotional Destabilization, Addictive Cognitive Capture, Trance or Policy Collapse, and Post-Acting-Out Aftermath. The companion engineering memorandum reformulated Active Inference in nonlinear stochastic state-space and finite-horizon control notation. The present memorandum combines these two ideas by proposing that regimes index different operating modes of the generative model.

The guiding principle is conservative:

> The optimization principles of Active Inference are preserved. The extension occurs in the latent-state architecture of the generative model.

Thus, ordinary variational inference about hidden states and ordinary Active Inference planning remain available. What changes is the predictive model under which hidden states, observations, and policies are evaluated.

1.1 The central mathematical idea

Let x_k denote a continuous hidden state and M_k denote a discrete regime variable. In the standard nonlinear state-space formulation, hidden states evolve according to

$$x_{k+1} = f(x_k, u_k, w_k), \tag{1}$$

where u_k is an action or intervention and w_k is process noise. The regime-dependent extension replaces (1) with

$$x_{k+1} = f_{M_k}(x_k, u_k, w_k). \tag{2}$$

Equation (2) is the minimal mathematical change. The same psychological variables may evolve differently depending on the active regime. For example, a social stressor may be absorbed and regulated in a recovery regime, but may amplify craving, euphoric recall, and rationalization in an addictive-cognitive-capture regime.

A fuller version of the same idea is that a regime does not merely select a drift function. It may index an entire generative-model configuration. We therefore introduce

$$\mathcal{G}_M = \big(f_M, h_M, p_M, \gamma_M, \lambda_M, \Pi_M\big), \tag{3}$$

where f_M denotes latent dynamics, h_M denotes an observation mapping, p_M denotes regime-specific priors and preference-related components, γ_M denotes policy precision, λ_M denotes interoceptive or social-affective precision, and Π_M denotes the policy repertoire available or salient in regime M.

The notation in (3) should not be read as requiring every component to vary in every application. It states what may vary. A parsimonious model may allow only f_M and Π_M to differ across regimes. A richer model may allow precision, observation reliability, and preferences to differ as well.

2 Relation to Standard Active Inference

The standard engineering form of the hidden-state and observation model is

$$x_{k+1} = f(x_k, u_k, w_k), \tag{4}$$
$$y_k = h(x_k, v_k), \tag{5}$$

where $x_k \in \mathbb{R}^n$, $y_k \in \mathbb{R}^m$, and w_k, v_k denote process and observation noise. Equivalently, one specifies the transition and observation densities

$$p(x_{k+1} \mid x_k, u_k), \tag{6}$$
$$p(y_k \mid x_k). \tag{7}$$

Variational free energy is used for inference about hidden states given observations already obtained. In a simplified static form, for hidden state x, observation y, and approximate posterior $q(x)$,

$$F[q] = D_{\mathrm{KL}}\big(q(x) \,\|\, p(x)\big) - \mathbb{E}_q\big[\log p(y \mid x)\big]. \tag{8}$$

Expected free energy is used for planning over possible future policies. For a policy π inducing future predictive distributions over observations, a useful decomposition is

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)}[-\log p_{\mathrm{pref}}(y_k)] - \sum_{k=t+1}^{t+H} I_k(\pi), \tag{9}$$

where $p_{\mathrm{pref}}(y_k)$ is a preference distribution and $I_k(\pi)$ is expected information gain about hidden states.

The regime-dependent framework developed here does not alter (8) or (9). Instead, it alters the generative model under which the predictive distributions are formed. This distinction is essential. The proposal is not that addiction requires a different optimization principle. The proposal is that addiction may be modeled as inference and planning under a generative model whose operating regime has changed.

3 Continuous Latent Affective-Cognitive State

The proposed first-pass continuous hidden state is

$$x_k = \begin{bmatrix} f_k \\ a_k \\ \rho_k \\ c_k \\ r_k \\ s_k \\ \ell_k \\ \gamma_k \\ \lambda_k \end{bmatrix} \in \mathbb{R}^9. \tag{10}$$

The components are defined in Table 1.

The vector in (10) should be understood as a modeling proposal, not as a final ontology of addiction. Its purpose is to identify variables that appear repeatedly in the phenomenology of relapse and recovery: fear, anger, loss of reality testing, craving, euphoric recall, shame, isolation, and changes in precision. Other applications may require additional coordinates, fewer coordinates, or different parameterizations.

Symbol	Interpretation
f_k	Fear or anxiety, interpreted as perceived threat intensity.
a_k	Anger or resentment, interpreted as perceived grievance, frustration, or injustice.
ρ_k	Reality-testing capacity; lower values correspond to impaired appraisal, self-deception, or reduced contact with consequences.
c_k	Craving or addictive action urge.
r_k	Euphoric recall or positive memory bias concerning prior acting out.
s_k	Shame, guilt, or self-condemnation.
ℓ_k	Perceived isolation or social disconnection.
γ_k	Policy precision or confidence in policies, often linked in the Active Inference literature to dopaminergic modulation.
λ_k	Interoceptive or social-affective precision, representing the weighting of bodily, affiliative, and social signals.

Table 1: Proposed first-pass latent state for a regime-dependent Active Inference model of the addictive cycle.

3.1 Latent rather than directly observed variables

The components of x_k are hidden variables. They are not assumed to be directly observed with perfect reliability. Observable quantities may include self-report, facial expression, heart rate variability, sleep disruption, behavioral markers, substance use, attendance at recovery meetings, journaling content, contact with social support, or ecological momentary assessment. These observations are generated from latent states through an observation model

$$y_k = h_{M_k}(x_k, v_k), \tag{11}$$

with corresponding density

$$p_{M_k}(y_k \mid x_k). \tag{12}$$

The subscript M_k allows the reliability or interpretation of observations to depend on regime. For example, self-report may be more reliable during Wise Mind or Recovery than during Addictive Cognitive Capture or Trance.

3.2 Precision variables as latent state components

The inclusion of γ_k and λ_k in (10) requires comment. In many Active Inference formulations, policy precision appears as a parameter controlling the sharpness of policy selection. Here it is included as a latent variable because confidence in available policies may itself evolve during the addictive cycle. In recovery, policy precision may be distributed across a broad repertoire of adaptive actions. During cognitive capture, confidence may increasingly attach to addictive policies. During trance or policy collapse, the effective precision of a narrow compulsive policy may dominate the policy landscape.

Similarly, λ_k represents the weighting of interoceptive and social-affective signals. Changes in the precision of bodily discomfort, social rejection, loneliness, shame, or craving may alter both state inference and future planning. Treating λ_k as latent keeps open the possibility that shifts in interoceptive or affiliative precision contribute to regime transitions.

4 Emotional Dysregulation as an Emergent Functional

Emotional dysregulation is not introduced as an additional coordinate of x_k. Instead, it is defined as an emergent functional of the latent state:

$$D_k = D(x_k; \theta), \tag{13}$$

where θ denotes parameters determining how latent variables combine into a macroscopic dysregulation index.

This definition is important conceptually. Dysregulation is not identical to fear, anger, craving, shame, or impaired reality testing. It summarizes the overall condition of the regulatory system. A person may experience fear without being globally dysregulated; another may have moderate fear, anger, isolation, and craving whose combined interaction produces a high-risk state. The functional $D(x_k; \theta)$ is intended to represent this system-level condition.

4.1 Linear parameterization

A simple interpretable first approximation is

$$D_k = \theta_0 + \theta_f f_k + \theta_a a_k - \theta_\rho \rho_k + \theta_c c_k + \theta_r r_k + \theta_s s_k + \theta_\ell \ell_k. \tag{14}$$

The negative sign in front of $\theta_\rho \rho_k$ reflects the assumption that reduced reality testing increases dysregulation. The coefficients in (14) may be constrained to be nonnegative for interpretability, although such constraints are not required mathematically.

4.2 Nonlinear parameterization

A more general representation is

$$D_k = \Phi(x_k; \theta), \tag{15}$$

where Φ may be a nonlinear function approximator, such as a generalized additive model, radial basis expansion, recurrent neural network, or multilayer perceptron. Nonlinear forms allow interactions such as

$$\frac{\partial^2 D}{\partial c\, \partial \rho} < 0, \tag{16}$$

which would express the possibility that craving becomes more destabilizing as reality testing declines.

The first mathematical formulation should not commit prematurely to a particular parameterization of D. Equation (13) defines the conceptual role of dysregulation; equations (14) and (15) provide possible implementations.

5 Discrete Regime Variable

Introduce a discrete hidden regime variable

$$M_k \in \mathcal{M} = \{M_1, M_2, M_3, M_4, M_5\}. \tag{17}$$

The five regimes are summarized in Table 2.

Regime	Name	Interpretation
M_1	Wise Mind / Recovery	Intact reality testing, regulatory capacity, broad policy space, access to recovery skills.
M_2	Emotional Destabilization	Fear, anger, emotional disturbance, declining regulation, increased vulnerability.
M_3	Addictive Cognitive Capture	Craving, euphoric recall, cognitive distortions, rituals, rationalizations, narrowing of attention.
M_4	Trance / Policy Collapse	Autopilot-like state, sharply restricted policy space, subjective loss of agency, acting out.
M_5	Post-Acting-Out Aftermath	Temporary relief followed by shame, despair, isolation, and renewed vulnerability.

Table 2: Five-regime structure motivated by the addictive cycle.

The regime variable is a higher-level hidden state of the generative model. It should not be confused with the Markov blanket itself. Nor should it be treated as a directly observed diagnostic category. It is a latent discrete variable inferred from observations, history, and the current latent affective-cognitive state.

5.1 Regimes as operating modes

The central interpretation is as follows:

> A psychological regime is an operating mode of the generative model.

This is stronger than saying that a regime is a behavioral stage. Within each regime, the same observed event may be assigned different meaning, latent variables may evolve according to different dynamics, and different policies may become salient or available. Thus, a regime is a configuration of inference and prediction, not merely a label placed on behavior after the fact.

6 Regime-Dependent Generative Model

A regime-dependent generative model is defined by the collection

$$\mathcal{G}_M = \big(f_M, h_M, p_M, \gamma_M, \lambda_M, \Pi_M\big), \tag{18}$$

where each component has the interpretation in Table 3.

This tuple is an architectural definition. It allows several levels of model complexity. At the most conservative level, only f_M varies across regimes. At an intermediate level, the transition dynamics and policy repertoire vary. At a richer level, observation reliability, precision parameters, and preference distributions may also differ.

6.1 Regime-dependent transition density

The deterministic-noise form

$$x_{k+1} = f_{M_k}(x_k, u_k, w_k) \tag{19}$$

Component	Interpretation
f_M	Regime-dependent latent dynamics.
h_M	Regime-dependent observation mapping or observation reliability.
p_M	Regime-dependent priors, preferences, or prior beliefs about hidden states and outcomes.
γ_M	Regime-dependent policy precision or inverse temperature.
λ_M	Regime-dependent interoceptive or social-affective precision.
Π_M	Regime-dependent policy repertoire or admissible policy set.

Table 3: Components of a regime-dependent generative model.

corresponds to the transition density

$$p(x_{k+1} \mid x_k, u_k, M_k). \tag{20}$$

For example, in an additive-noise model,

$$x_{k+1} = F_{M_k}(x_k, u_k) + w_k, \qquad w_k \sim \mathcal{N}(0, Q_{M_k}), \tag{21}$$

so that

$$p(x_{k+1} \mid x_k, u_k, M_k) = \mathcal{N}\big(x_{k+1}; F_{M_k}(x_k, u_k), Q_{M_k}\big). \tag{22}$$

The covariance Q_{M_k} may itself be regime dependent. This permits greater stochastic volatility during destabilization or aftermath than during stable recovery.

6.2 Local linear approximation

For analysis or identification, one may linearize the regime-dependent dynamics around a nominal trajectory:

$$x_{k+1} \approx A_{M_k} x_k + B_{M_k} u_k + b_{M_k} + w_k. \tag{23}$$

This form makes clear that each regime may possess different gains, offsets, and noise structure. For example, the gain from a stressor to craving may be small in M_1 and large in M_3. The gain from a recovery action to reality testing may be large in M_1 or M_2, but attenuated in M_4.

6.3 Observation model

A regime-dependent observation model has the form

$$y_k = h_{M_k}(x_k, v_k), \tag{24}$$

with density

$$p(y_k \mid x_k, M_k). \tag{25}$$

In additive Gaussian form,

$$y_k = H_{M_k}(x_k) + v_k, \qquad v_k \sim \mathcal{N}(0, R_{M_k}). \tag{26}$$

The dependence of R_{M_k} on regime permits the reliability of measurements or self-report to vary across operating modes. In recovery, verbal reports of craving and intention may be relatively informative. In trance or policy collapse, self-report may become delayed, confabulated, or unavailable.

7 Regime Transition Model

The regime transition model is

$$P(M_{k+1} = j \mid M_k = i, x_k, u_k), \qquad i, j \in \{1, \ldots, 5\}. \tag{27}$$

Equivalently, define a transition matrix depending on state and action:

$$T_{ij}(x_k, u_k) = P(M_{k+1} = M_j \mid M_k = M_i, x_k, u_k), \tag{28}$$

with

$$\sum_{j=1}^{5} T_{ij}(x_k, u_k) = 1, \qquad T_{ij}(x_k, u_k) \geq 0. \tag{29}$$

7.1 Dysregulation-driven transitions

The dysregulation functional is expected to play a central role in regime transitions. A simple threshold formulation introduces a Skills Breakdown Point D_{SBP} and an Autopilot threshold D_{AP}, with

$$D_{\mathrm{AP}} > D_{\mathrm{SBP}}. \tag{30}$$

The event

$$D(x_k; \theta) > D_{\mathrm{SBP}} \tag{31}$$

indicates entry into a transition region where ordinary recovery skills become less accessible or less likely to be selected. The event

$$D(x_k; \theta) > D_{\mathrm{AP}} \tag{32}$$

indicates high risk of Trance or Policy Collapse.

A smooth alternative uses multinomial logistic transition probabilities:

$$T_{ij}(x_k, u_k) = \frac{\exp\{\alpha_{ij} + \beta_{ij} D(x_k; \theta) + \eta_{ij}^{\mathsf{T}} u_k\}}{\sum_{\ell=1}^{5} \exp\{\alpha_{i\ell} + \beta_{i\ell} D(x_k; \theta) + \eta_{i\ell}^{\mathsf{T}} u_k\}}. \tag{33}$$

This form permits thresholds to be approximated continuously and allows interventions u_k to shift transition probabilities.

7.2 Dominant relapse pathway

The dominant relapse pathway is represented as

$$M_1 \to M_2 \to M_3 \to M_4 \to M_5. \tag{34}$$

In terms of transition probabilities, (34) means that under increasing dysregulation and absent effective intervention,

$$T_{12}(x_k, u_k) \text{ increases as } D(x_k; \theta) \text{ rises,} \tag{35}$$

$$T_{23}(x_k, u_k) \text{ increases as craving and euphoric recall dominate,} \tag{36}$$

$$T_{34}(x_k, u_k) \text{ increases as the policy repertoire collapses,} \tag{37}$$

$$T_{45}(x_k, u_k) \text{ increases after acting out has occurred.} \tag{38}$$

The pathway in (34) is not deterministic. It is a high-probability sequence under particular latent and contextual conditions.

7.3 Recovery-oriented transitions

Recovery-oriented transitions must also be allowed. Examples include

$$M_2 \to M_1, \qquad M_3 \to M_2, \qquad M_3 \to M_1. \tag{39}$$

These may be driven by effective interventions u_k, such as reaching out to a sponsor, attending a meeting, changing environment, using DBT or CBT skills, prayer or meditation, delay, sleep, exercise, or explicit reality testing.

The aftermath regime also feeds back into vulnerability:

$$M_5 \to M_2, \qquad M_5 \to M_3. \tag{40}$$

This represents the clinically important possibility that shame and isolation following acting out reinitiate emotional destabilization or cognitive capture.

8 The Hybrid Generative Process

The complete regime-dependent generative process may be written as

$$M_{k+1} \sim P(M_{k+1} \mid M_k, x_k, u_k), \tag{41}$$

$$x_{k+1} \sim p(x_{k+1} \mid x_k, u_k, M_k), \tag{42}$$

$$y_k \sim p(y_k \mid x_k, M_k). \tag{43}$$

Depending on modeling convention, x_{k+1} may be conditioned on M_k or M_{k+1}. Conditioning on M_k emphasizes that the current operating mode generates the next continuous state. Conditioning on M_{k+1} emphasizes that the newly entered regime determines the next state update. Both conventions are possible; the choice should be stated explicitly in any implementation.

One convenient alternative is

$$M_{k+1} \sim P(M_{k+1} \mid M_k, x_k, u_k), \tag{44}$$

$$x_{k+1} \sim p(x_{k+1} \mid x_k, u_k, M_{k+1}), \tag{45}$$

$$y_{k+1} \sim p(y_{k+1} \mid x_{k+1}, M_{k+1}). \tag{46}$$

This convention is often natural for forward prediction: first infer the next regime, then propagate the continuous state under that regime.

8.1 Graphical dependency summary

The proposed dependency structure can be summarized as

$$x_k \longrightarrow D(x_k;\theta) \longrightarrow M_{k+1} \longrightarrow \mathcal{G}_{M_{k+1}} \longrightarrow x_{k+1}, y_{k+1}. \tag{47}$$

A more compact conceptual hierarchy is

$$x_k \longrightarrow D_k \longrightarrow M_k \longrightarrow \mathcal{G}_{M_k}. \tag{48}$$

This hierarchy expresses the bridge between addiction phenomenology and Active Inference: continuous latent affective-cognitive states generate an emergent dysregulation index; dysregulation shapes regime probabilities; regimes index generative-model configurations; and those configurations determine future latent dynamics and observations.

9 Regime-Specific Qualitative Parameterization

This section gives a qualitative first-pass parameterization of each regime. It is not intended as a final quantitative model. It identifies which components of $\mathcal{G}_M$ may plausibly differ across regimes.

9.1 M_1: Wise Mind / Recovery

In M_1, reality testing is relatively high, dysregulation is low or moderate, and the policy repertoire is broad. A local linear approximation might have stabilizing dynamics:

$$\rho_{k+1} - \rho_k > 0 \quad \text{under recovery actions}, \qquad c_{k+1} - c_k < 0 \quad \text{under recovery actions}. \tag{49}$$

The policy repertoire Π_{M_1} includes adaptive actions such as delay, reaching out, cognitive reappraisal, meeting attendance, spiritual practice, environmental change, and self-care. Policy precision is not necessarily low; rather, it is distributed over adaptive policies rather than concentrated on addictive action.

9.2 M_2: Emotional Destabilization

In M_2, fear, anger, or distress increase, while regulatory capacity begins to decline. Dynamics may amplify affective variables:

$$f_{k+1} \uparrow, \qquad a_{k+1} \uparrow, \qquad \rho_{k+1} \downarrow . \tag{50}$$

The policy space may remain broad, but access to it becomes less reliable. Interventions may still be highly effective in this regime, making transitions back to M_1 clinically important.

9.3 M_3: Addictive Cognitive Capture

In M_3, craving, euphoric recall, rationalization, and attentional narrowing become dominant. The dynamics may include positive feedback between craving and euphoric recall:

$$c_{k+1} = F_c(c_k, r_k, \rho_k, u_k, w_k), \qquad r_{k+1} = F_r(r_k, c_k, \rho_k, u_k, w_k), \tag{51}$$

with

$$\frac{\partial F_c}{\partial r} > 0, \qquad \frac{\partial F_r}{\partial c} > 0. \tag{52}$$

Declining reality testing may increase the influence of euphoric recall on craving:

$$\frac{\partial^2 F_c}{\partial r \, \partial \rho} < 0. \tag{53}$$

This expresses the idea that positive memory bias becomes more destabilizing as appraisal weakens.

9.4 M_4: Trance / Policy Collapse

In M_4, the policy repertoire is sharply restricted. A simple mathematical representation is

$$\Pi_{M_4} \subsetneq \Pi_{M_3} \subsetneq \Pi_{M_1}. \tag{54}$$

The effective posterior over policies may become highly concentrated on a narrow addictive policy set. In this memorandum the details of policy selection are deferred to Memorandum IIIB, but the generative-model architecture already permits collapse of the admissible or salient policy repertoire.

9.5 M_5: Post-Acting-Out Aftermath

In M_5, immediate craving may temporarily decrease, but shame and isolation may increase:

$$c_{k+1} \downarrow, \qquad s_{k+1} \uparrow, \qquad \ell_{k+1} \uparrow . \tag{55}$$

This regime is not simply the end of the cycle. It may create conditions for renewed destabilization or cognitive capture through the transitions in (40).

10 Identification and Data Considerations

The framework can be estimated at several levels of ambition. A minimal model might specify $D(x_k;\theta)$ and transition probabilities by expert judgment. A data-driven model might infer both the latent state and the regime sequence from longitudinal observations.

Given observations $y_{1:T}$ and actions or interventions $u_{1:T}$, the latent variables are

$$x_{1:T}, \qquad M_{1:T}. \tag{56}$$

The joint probability under a regime-dependent model may be written as

$$\begin{aligned} p(x_{1:T}, M_{1:T}, y_{1:T} \,|\, u_{1:T}) = {} & p(M_1)p(x_1 \,|\, M_1)p(y_1 \,|\, x_1, M_1) \\ & \times \prod_{k=1}^{T-1} P(M_{k+1} \,|\, M_k, x_k, u_k) \\ & \times p(x_{k+1} \,|\, x_k, u_k, M_{k+1})p(y_{k+1} \,|\, x_{k+1}, M_{k+1}). \end{aligned} \tag{57}$$

Approximate inference may then be performed using variational methods, particle filtering, expectation-maximization, sequential Monte Carlo, or amortized inference.

Potential data sources include self-report, ecological momentary assessment, daily inventories, relapse records, journaling, wearable physiology, sleep measures, geolocation context, social-contact patterns, and treatment engagement. The framework does not require all such data; it provides a place for them in a coherent generative model.

11 Open Modeling Choices

Several choices should remain open in the first generation of this framework:

1. What is the optimal latent state representation x_k?
2. Should fear and anger remain separate coordinates, or should they be combined into a broader threat or resentment variable?
3. Should γ_k and λ_k be treated as latent states, regime parameters, or both?
4. What parameterization of $D(x_k;\theta)$ is most interpretable and empirically identifiable?
5. Should regime transitions be threshold-based, softmax-based, semi-Markov, or learned using a more flexible model?
6. Which components of $\mathcal{G}_M$ should vary across regimes in the minimal useful model?
7. Can longitudinal behavioral, physiological, or journaling data identify the transition probabilities $P(M_{k+1} \,|\, M_k, x_k, u_k)$?

8. Should dysregulation remain only a transition-driving functional, or should it eventually enter preferences or expected free energy explicitly?

These questions are not defects in the formulation. They identify the empirical and mathematical work required to turn the architecture into an estimable model.

12 Summary

This memorandum has proposed a minimal hybrid extension of the Active Inference generative model. The central definitions are:

$$x_k = (f_k, a_k, \rho_k, c_k, r_k, s_k, \ell_k, \gamma_k, \lambda_k)^{\mathsf{T}}, \tag{58}$$

$$D_k = D(x_k; \theta), \tag{59}$$

$$M_k \in \{M_1, M_2, M_3, M_4, M_5\}, \tag{60}$$

$$\mathcal{G}_M = (f_M, h_M, p_M, \gamma_M, \lambda_M, \Pi_M), \tag{61}$$

$$P(M_{k+1} \,|\, M_k, x_k, u_k). \tag{62}$$

The essential hierarchy is

$$x_k \longrightarrow D(x_k; \theta) \longrightarrow M_k \longrightarrow \mathcal{G}_{M_k}. \tag{63}$$

The scientific claim is deliberately modest. Addiction is not modeled as a failure of free-energy minimization. Rather, addiction and recovery are modeled as inference and planning under regime-dependent generative models whose latent dynamics, precision structure, and policy landscape evolve with emotional dysregulation. The next memorandum, IIIB, uses the architecture defined here to show how standard Active Inference planning and policy selection proceed when future trajectories include both continuous latent states and discrete regimes.

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